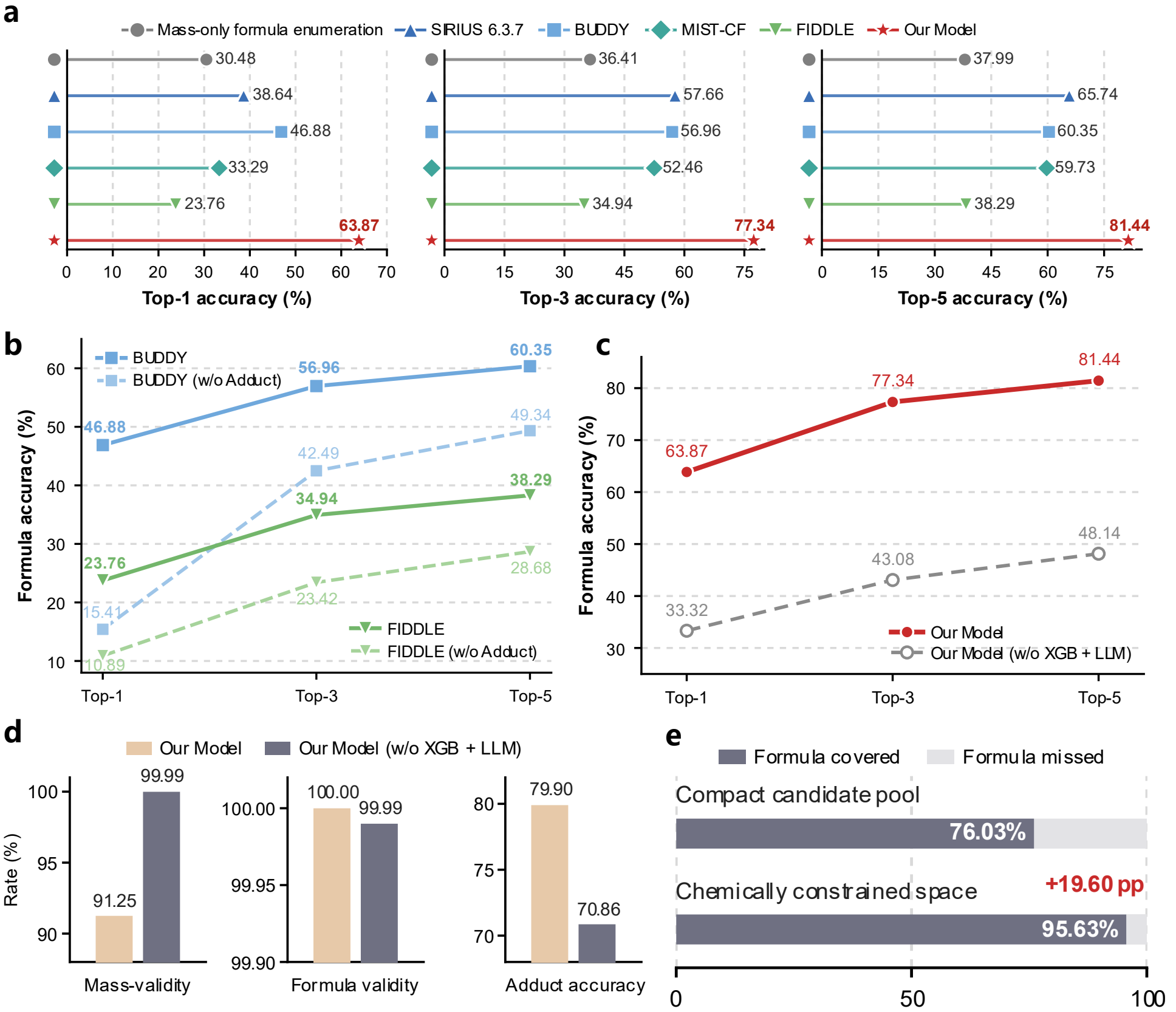

a
Mass-only formula enumeration
SIRIUS 6.3.7
BUDDY
MIST-CF
FIDDLE
Our Model
30.48
38.64
46.88
33.29
23.76
63.87
Top-1 accuracy (%)
36.41
57.66
56.96
52.46
34.94
77.34
Top-3 accuracy (%)
37.99
65.74
60.35
59.73
38.29
81.44
Top-5 accuracy (%)
b
BUDDY
BUDDY (w/o Adduct)
FIDDLE
FIDDLE (w/o Adduct)
Formula accuracy (%)
Top-1
Top-3
Top-5
c
Our Model
Our Model (w/o XGB + LLM)
Formula accuracy (%)
d
Our Model
Our Model (w/o XGB + LLM)
Rate (%)
Mass-validity
Formula validity
Adduct accuracy
e
Formula covered
Formula missed
Compact candidate pool
76.03%
Chemically constrained space
+19.60 pp
95.63%